Do Now · retrieval

Quicksort: tracing and implementation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

After partitioning [8,1,7,3,6] around pivot 6, where is the pivot?

Q2

Why does recursive quicksort need a base case?

Q3

What is the worst-case time complexity of merge sort?

Q4

If the pivot finishes at index 3, give left and right recursive bounds in an array 0..6.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Stop small subarrays — When low>=high there are zero or one elements to sort: return immediately.

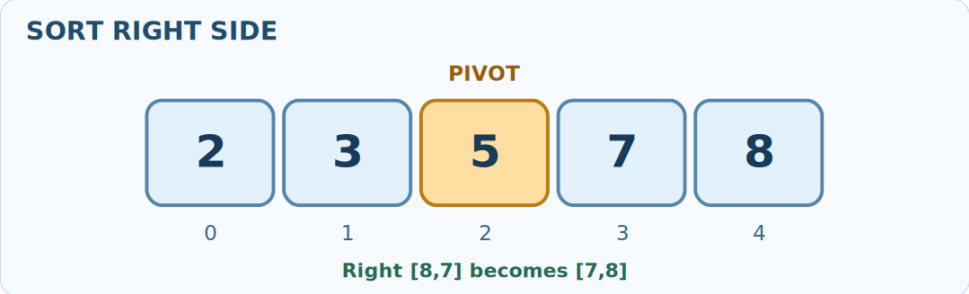
Core idea 2

Partition then divide — Partition places the pivot at its final index p. Do not recursively include index p.

Core idea 3

Sort two sides independently — Call QuickSort on low..p−1 and p+1..high. The pivot itself stays fixed.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

quicksort — Recursive sort using partitioning around a pivot.

base case — Condition ending recursion.

partition index — Final index occupied by the pivot.

worst case — Input arrangement causing the most expensive run.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

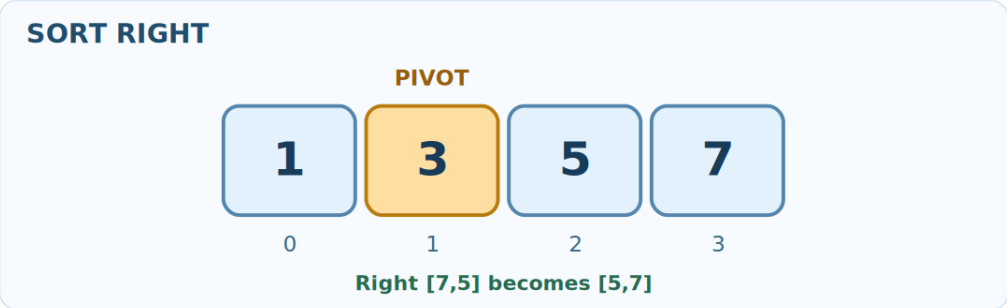
Problem

Trace recursive quicksort on [5,1,7,3] (last pivot).

Follow the worked method

1. Predict the next state — Partition around 3, moving 1 to the left and placing pivot 3 at index 1.
2. Reveal and explain — Left segment [1] is already sorted; sort only the right segment [7,5].
3. Reveal and explain — The right recursion produces [5,7]; final array [1,3,5,7].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next state
Partition around pivot 4, producing [1,2,4,6,5].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Why is if(low>=high) return; required in a recursive C# quicksort?

Question 2

Use last-element Lomuto quicksort on [4,1,6,2]. Give the first partition and the final sorted array.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Describe the average and worst-case time complexity of quicksort, why these can differ, and one implementation trade-off relative to merge sort.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

After partition, the recursive calls should include the pivot index again.

Use key vocabulary from memory

1. Explain quicksort in your own words.

2. Explain base case in your own words.

3. Explain partition index in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.